Activity

Measuring the speed of light using the microwave oven

Nowadays the microwave oven is very commonly used to heat the food items. Micro waves of wavelengths 1 mm to 30 cm are produced in these ovens. Such waves form the standing waves between the interior walls of the oven. It is interesting to note that the speed of light can be measured using micro wave oven.



(a) (b) (c) (d)

We studied about the standing waves in XI physics, Volume 2, Unit 11. The standing waves have nodes and antinodes at fixed points. At node point, the amplitude of the wave is zero and at antinodes point, the amplitude is maximum. In other words, the maximal energy of microwaves is located at antinode points. When we keep some food items like chappathi or choclate (after removing the rotating platform) inside the oven, we can notice that at antinode locations, chappathi will be burnt more than other locations. It is shown in the Figure (c) and (d). The distance between two successive burnt spots will give the wavelength of microwave. The frequency of microwave is printed in the panel of oven. By knowing wavelength and frequency of microwaves, using the formula $v\lambda = c$, we can calculate the speed of light *c*.

Infrared radiation

It is produced from hot bodies (also known as heat waves) and also when the molecules undergo rotational and vibrational transitions. The wavelength range is 8×10^{-7} m to 5×10^{-3} m and frequency range are 4×10^{14} Hz to 6×10^{10} Hz. It provides electrical energy to satellites by means of solar cells. It is used to produce dehydrated fruits, in green houses to keep the plants warm, heat therapy for muscular pain or sprain, TV remote as a signal carrier, to look through haze fog or mist and used in night vision or infrared photography.

Visible light

It is produced by incandescent bodies and also it is radiated by excited atoms in gases. The wavelength range is 4×10^{-7} m to 7×10^{-7} m and frequency range are 7×10^{14} Hz to 4×10^{14} Hz. It obeys the laws of reflection, refraction, interference, diffraction, polarization, photo-electric effect and photographic action. It can be used to study the structure of molecules, arrangement of electrons in external shells of atoms and sensation of our eyes.

Ultraviolet radiation

It is produced by Sun, arc and ionized gases. The wavelength range is 6×10^{-10} m to 4×10^{-7} m and frequency range are 5×10^{17} Hz to 7×10^{14} Hz. It has less penetrating power. It can be absorbed by atmospheric ozone and harmful to human body. It is used to destroy bacteria, sterilizing the surgical instruments, burglar alarm, detect the invisible writing, finger prints and also in the study of molecular structure.

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